# OSINT PRACTICE – Collection and analysis of public information

ESCUELA COLOMBIANA DE INGENIERÍA JULIO GARAVITO

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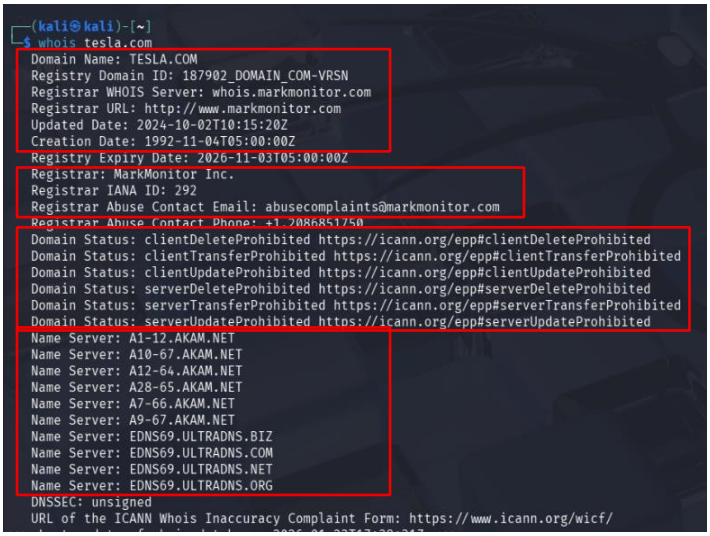
**26/01/26**

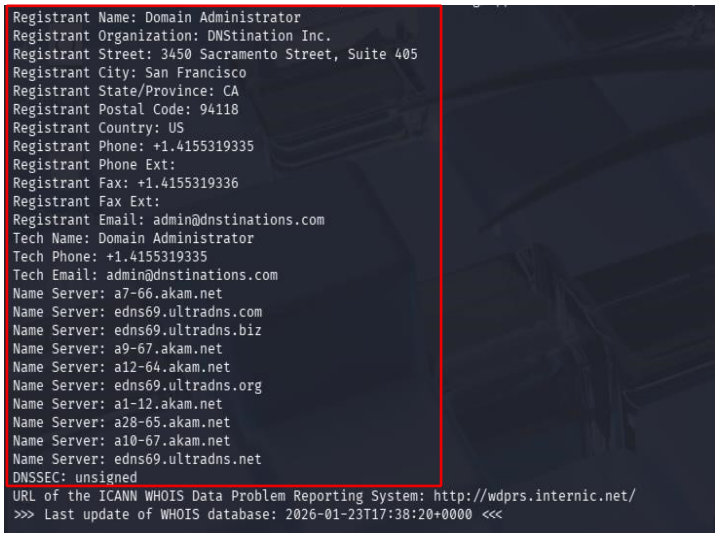
**Techniques and tools used:**

The analysis was carried out in a virtualized environment with VMware, using a Kali Linux virtual machine, which allowed working in an isolated and controlled way. ExifTool was used to extract metadata from PDFs, DOCXs, images and presentations, identifying relevant information such as authorship, dates, software used and geolocation using Google Maps, without altering the original files. As a complement, online platforms were used for cross-checking metadata such as metadata2go.com. Finally, the information collected was correlated using Maltego Community Edition, allowing the visualization of relationships between people, organizations and locations, and the export of the graph as evidence for the report.

## 1. Part 1: Passive Collection on a domain or person.

- Because of the love of vehicles and their technological advances, we decided to choose the tesla.com domain.

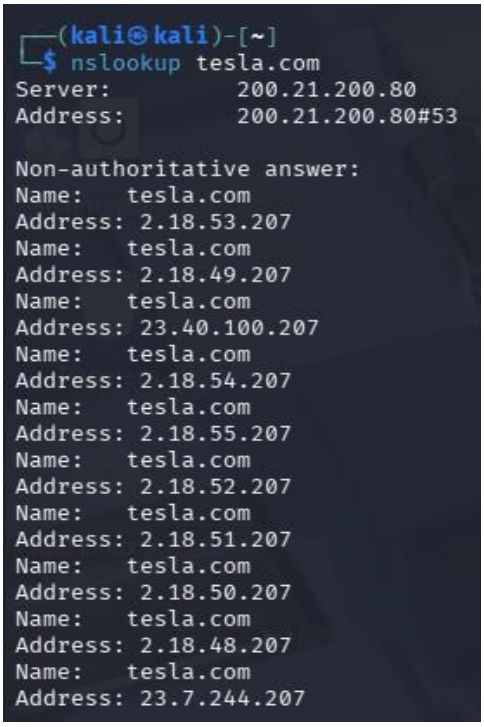


In the images we can see that the domain "tesla.com" was created in 1992 being 33 years old, a domain with enough time gives the confidence that it is a registered trademark, it is not a temporary domain; is registered by MarkMonitor Inc which is a registrar of large corporations it carries out services such as brand protection, anti-phishing and anti hijacking, We observe statuses such as:

- "clientDeleteProhibited"
- "clientTransferProhibited"
- "clientUpdateProhibited"
- "serverDeleteProhibited"
- "serverTransferProhibited"
- "serverUpdateProhibited"

this means that the domain is well protected to prevent domain hijacking by prohibiting domain deletion, unauthorized transfers or unauthorized updates, in terms of DNS infrastructure we can see 2 DNS servers detected *.akam.net which is Akamai and offers DDoS protection and *.ultradns.* which is a highly available DNS resistant to attacks; Finally, we see the registrant's data such as their location, contact email, zip code, phone, this company is an intermediary of tesla fulfilling the function of hiding Tesla's information, MarkMonitor registers and protects the domain with ICANN; DNStination is listed as the administrative holder to hide and manage Tesla's actual data.




By performing the nslookup tesla.com command we can first see the "Nonauthoritative answer" which means that the DNS server is not the official server of tesla.com, in addition, 10 different IPv4 addresses are resolved for the same domain, which indicates that it does not point to a single server and the traffic is distributed to different servers indicating the use of a DNS-based load balancing redirecting the user to the nearest node or available.

```
┌──(kali㉿kali)-[~]
└─$ dig tesla.com A

; <<>> DiG 9.20.15-2-Debian <<>> tesla.com A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 51572
;; flags: qr rd ra; QUERY: 1, ANSWER: 10, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;tesla.com.                     IN      A

;; ANSWER SECTION:
tesla.com.              300     IN      A       2.18.51.207
tesla.com.              300     IN      A       23.40.100.207
tesla.com.              300     IN      A       2.18.50.207
tesla.com.              300     IN      A       2.18.54.207
tesla.com.              300     IN      A       2.18.48.207
tesla.com.              300     IN      A       2.18.52.207
tesla.com.              300     IN      A       2.18.53.207
tesla.com.              300     IN      A       2.18.55.207
tesla.com.              300     IN      A       2.18.49.207
tesla.com.              300     IN      A       23.7.244.207

;; Query time: 43 msec
;; SERVER: 200.21.200.80#53(200.21.200.80) (UDP)
;; WHEN: Fri Jan 23 14:11:00 -05 2026
;; MSG SIZE  rcvd: 198
```

```
┌──(kali㉿kali)-[~]
└─$ dig tesla.com MX

; <<>> DiG 9.20.15-2-Debian <<>> tesla.com MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 26176
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;tesla.com.                     IN      MX

;; ANSWER SECTION:
tesla.com.              300     IN      MX      10 tesla-com.mail.protection.outlook.com.

;; Query time: 47 msec
;; SERVER: 200.21.200.80#53(200.21.200.80) (UDP)
;; WHEN: Fri Jan 23 15:33:56 -05 2026
;; MSG SIZE  rcvd: 88
```

```
┌──(kali㉿kali)-[~]
└─$ dig tesla.com TXT
;; Truncated, retrying in TCP mode.
;; Connection to 200.21.200.80#53(200.21.200.80) for tesla.com failed: host unreachable.
;; no servers could be reached
;; Connection to 200.21.200.80#53(200.21.200.80) for tesla.com failed: host unreachable.
;; no servers could be reached
;; Connection to 200.21.200.80#53(200.21.200.80) for tesla.com failed: timed out.
;; Connection to 200.21.200.10#53(200.21.200.10) for tesla.com failed: host unreachable.
;; no servers could be reached
```

```
┌──(kali㉿kali)-[~]
└─$ theHarvester -d tesla.com -b bing,duckduckgo,yahoo,dnsdumpster -l 200
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*******************************************************************
*                                                                 *
* | |_| |_   ___    ^  /\__ _ _ ____   _____  ___| |_ ___ _ __  *
* | __|  _ \ / _ \  / /_/ / _` | '__\ \ / / _ \/ __| __/ _ \ '__| *
* | |_| | | |  __/ / __  / (_| | |   \ V /  __/\__ \ ||  __/ |    *
*  \__|_| |_|\___| v /_/ \__,_|_|    \_/ \___||___/\__\___|_|    *
*                                                                 *
* theHarvester 4.8.2                                              *
* Coded by Christian Martorella                                   *
* Edge-Security Research                                          *
* cmartorella@edge-security.com                                   *
*                                                                 *
*******************************************************************

[*] Target: tesla.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for DNSDumpster.
        Searching 0 results.
[*] Searching Bing.
[*] Searching Yahoo.
[*] Searching Duckduckgo.

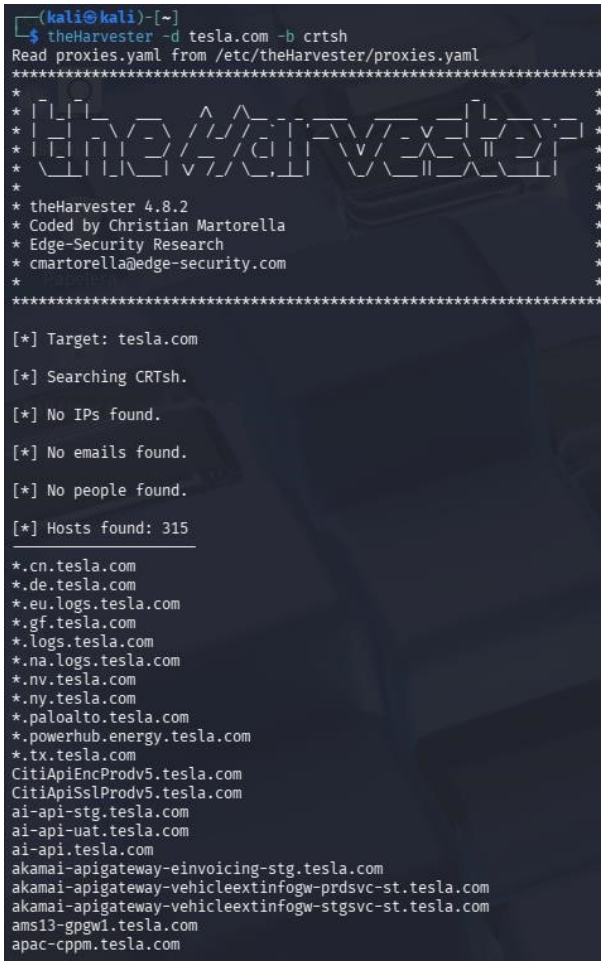
[*] No IPs found.

[*] No emails found.

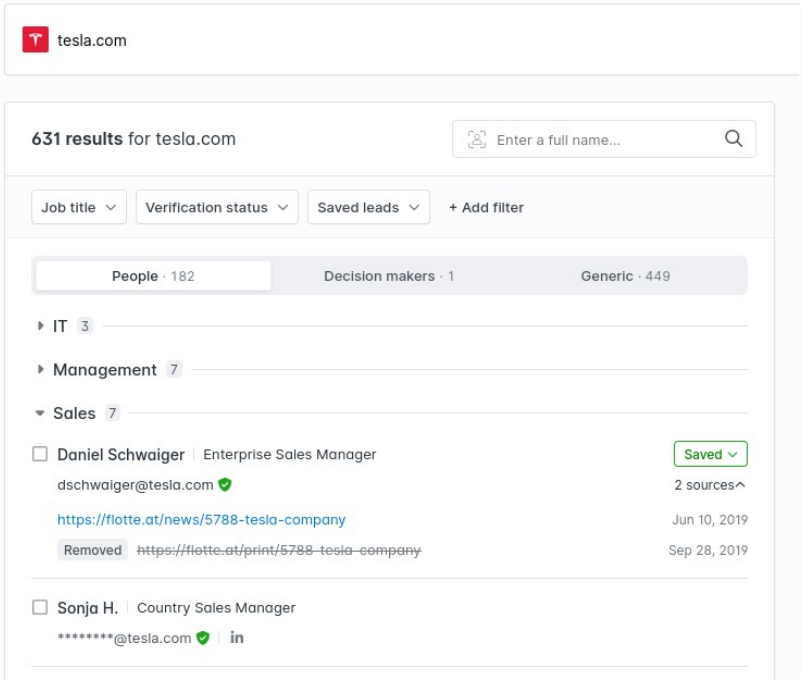
[*] No people found.

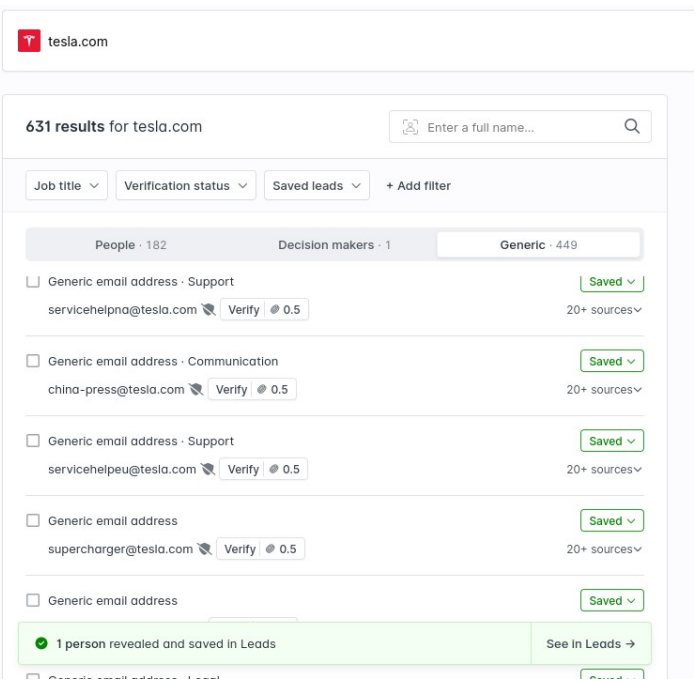
[*] Hosts found: 14
---------------------
2Fdigitalassets-shop.tesla.com
2Fdigitalassets.tesla.com
2Fir.tesla.com
accounts.tesla.com
assets-ir.tesla.com
billing.tesla.com
developer.tesla.com
digitalassets.tesla.com
epc.tesla.com
inside.tesla.com
ir.tesla.com
profile.tesla.com
repair.tesla.com
shop.tesla.com
```

A focused search on tesla.com using Bing, Yahoo, and DuckDuckGo returned no emails or IP addresses, which is consistent with data protection practices and the use of CDNs in large corporations. However, 14 relevant public subdomains were identified, including accounts.tesla.com, developer.tesla.com, billing.tesla.com, shop.tesla.com, repair.tesla.com and ir.tesla.com that show legitimately exposed services and make up the public surface of the domain. No DNSDumpster results were obtained due to lack of API key, and no social media profiles were identified due to current limitations of the module.




Enumeration using Certificate Transparency (crt.sh) for tesla.com identified 315 public subdomains, without revealing emails, IP addresses, or associated persons, which is consistent with the use of CDNs and strict privacy policies. The results show a wide public area that includes authentication services, APIs, regional infrastructure, staging/production environments, internal platforms, energy services and technical portals, evidencing the complexity and global distribution of Tesla's infrastructure. These subdomains come exclusively from public TLS certificate registries and do not involve vulnerabilities or unauthorized access.



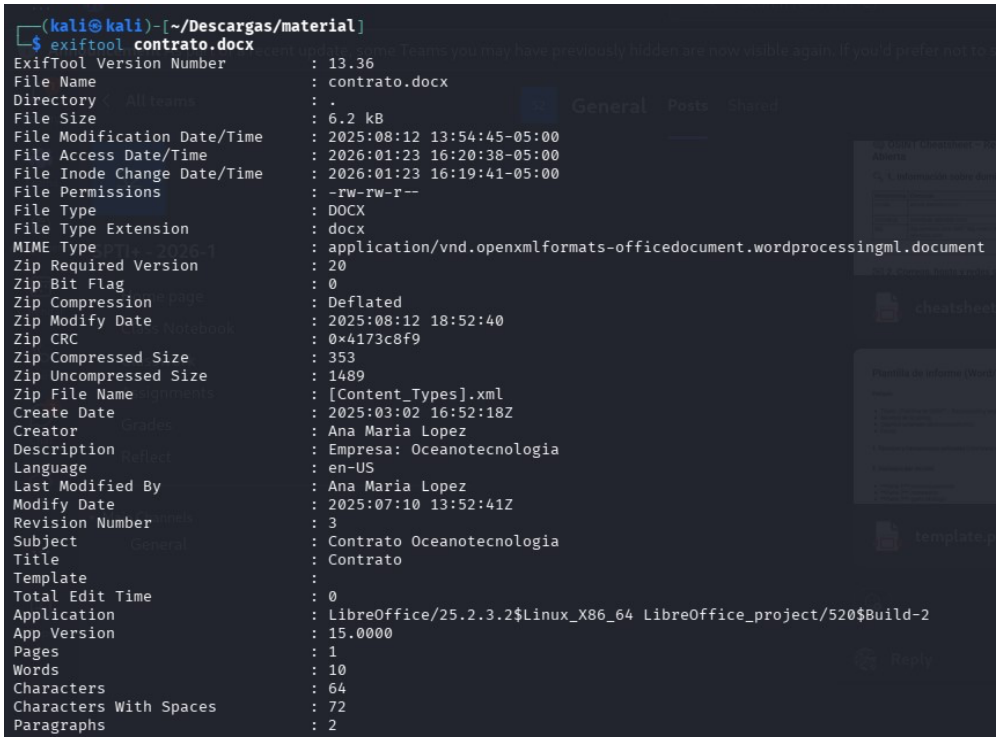

We can see that for the domain tesla.com we found 631 results, for people it uses first and last initial.

## 2. Part 2: Metadata analysis.

The files analyzed correspond to documents and digital native images (DOCX, PDF, PPTX, JPG and PNG), with no signs of scanning. Most of them identify persistent metadata such as names of people, dates of creation and modification, language and software used. The use of LibreOffice 25.2.3.2 predominates over Linux x86_64 office documents, which allows inferring a common work environment. None of the files, except for an image, contain encryption, active scripts, or protection mechanisms. Temporal information is consistent and useful for forensic chronology, and in several cases personal and organizational data are exposed, which increases the level of sensitivity of the whole.

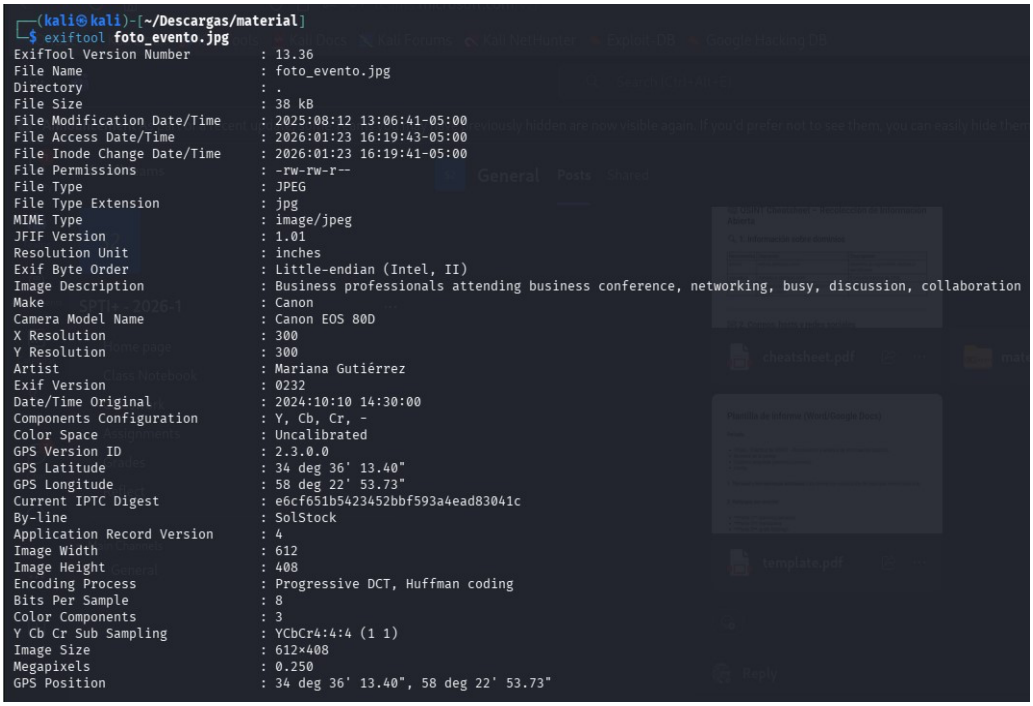
```
┌──(kali㉿kali)-[~/Descargas/material]
└─$ exiftool informe_proyecto.pdf
ExifTool Version Number         : 13.36
File Name                       : informe_proyecto.
pdf
Directory                       : .
File Size                       : 22 kB
File Modification Date/Time     : 2025:08:12 12:59:
36-05:00
File Access Date/Time           : 2026:01:23 16:19:
43-05:00
File Inode Change Date/Time     : 2026:01:23 16:19:
41-05:00
File Permissions                : -rw-rw-r--
File Type                       : PDF
File Type Extension             : pdf
MIME Type                       : application/pdf
Linearized                      : No
Page Count                      : 1
Page Mode                       : UseOutlines
Language                        : en-US
Tagged PDF                      : Yes
XMP Toolkit                     : Image::ExifTool 1
3.25
Contributor                     : Luis Fernandez
Date                            : 2025:08:12 12:00:
54-05:00
Format                          : application/pdf
Title                           : Informe de Proyec
to
PDF Version                     : 1.7
Producer                        : LibreOffice 25.2.
3.2 (X86_64) / LibreOffice Community
Create Date                     : 2024:11:14 12:00:
00
Creator Tool                    : Writer
Metadata Date                   : 2025:08:12 12:00:
54-05:00
Modify Date                     : 2025:08:12 12:00:
54-05:00
Creator                         : Writer
```

This file exposes the name Luis Fernandez in the Contributor field, a highly sensitive piece of information, since it directly identifies a person associated with the document. It includes clear dates of creation and modification, relevant to forensic chronology. The UTC-5 time zone limits the location to regions such as Colombia, Peru or Ecuador, without allowing you to confirm a specific country, as well as restricting our geographical field increases the chances of easier social engineering.



```
┌──(kali㉿kali)-[~/Descargas/material]
└─$ exiftool contrato.docx
ExifTool Version Number         : 13.36
File Name                       : contrato.docx
Directory                       : .
File Size                       : 6.2 kB
File Modification Date/Time     : 2025:08:12 13:54:45-05:00
File Access Date/Time           : 2026:01:23 16:20:38-05:00
File Inode Change Date/Time     : 2026:01:23 16:19:41-05:00
File Permissions                : -rw-rw-r--
File Type                       : DOCX
File Type Extension             : docx
MIME Type                       : application/vnd.openxmlformats-officedocument.wordprocessingml.document
Zip Required Version            : 20
Zip Bit Flag                    : 0
Zip Compression                 : Deflated
Zip Modify Date                 : 2025:08:12 18:52:40
Zip CRC                         : 0x4173c8f9
Zip Compressed Size             : 353
Zip Uncompressed Size           : 1489
Zip File Name                   : [Content_Types].xml
Create Date                     : 2025:03:02 16:52:18Z
Creator                         : Ana Maria Lopez
Description                     : Empresa: Oceanotecnologia
Language                        : en-US
Last Modified By                : Ana Maria Lopez
Modify Date                     : 2025:07:10 13:52:41Z
Revision Number                 : 3
Subject                         : Contrato Oceanotecnologia
Title                           : Contrato
Template                        :
Total Edit Time                 : 0
Application                     : LibreOffice/25.2.3.2$Linux_X86_64 LibreOffice_project/520$Build-2
App Version                     : 15.0000
Pages                           : 1
Words                           : 10
Characters                      : 64
Characters With Spaces          : 72
Paragraphs                      : 2
```

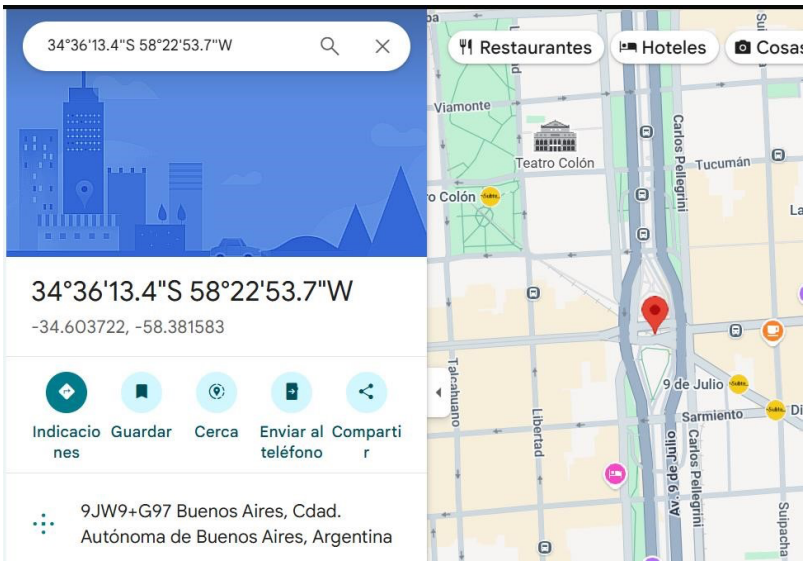
Highly sensitive personal data such as Ana Maria Lopez is identified in both Creator and Last Modified By, directly linking a person to the creation and editing of the file. In addition, the Oceanotecnologia entity is explicitly mentioned, establishing a person-company relationship, thus opening an important gap for easier social engineering execution by reducing the population to a single company with a specific name.



```
┌──(kali㉿kali)-[~/Descargas/material]
└─$ exiftool foto_evento.jpg
ExifTool Version Number         : 13.36
File Name                       : foto_evento.jpg
Directory                       : .
File Size                       : 38 kB
File Modification Date/Time     : 2025:08:12 13:06:41-05:00
File Access Date/Time           : 2026:01:23 16:19:43-05:00
File Inode Change Date/Time     : 2026:01:23 16:19:41-05:00
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : inches
Exif Byte Order                 : Little-endian (Intel, II)
Image Description               : Business professionals attending business conference, networking, busy, discussion, collaboration
Make                            : Canon
Camera Model Name               : Canon EOS 80D
X Resolution                    : 300
Y Resolution                    : 300
Artist                          : Mariana Gutiérrez
Exif Version                    : 0232
Date/Time Original              : 2024:10:10 14:30:00
Components Configuration        : Y, Cb, Cr, -
Color Space                     : Uncalibrated
GPS Version ID                  : 2.3.0.0
GPS Latitude                    : 34 deg 36' 13.40"
GPS Longitude                   : 58 deg 22' 53.73"
Current IPTC Digest             : e6cf651b5423452bbf593a4ead83041c
By-line                         : SolStock
Application Record Version      : 4
Image Width                     : 612
Image Height                    : 408
Encoding Process                : Progressive DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Image Size                      : 612x408
Megapixels                      : 0.250
GPS Position                    : 34 deg 36' 13.40", 58 deg 22' 53.73"
```

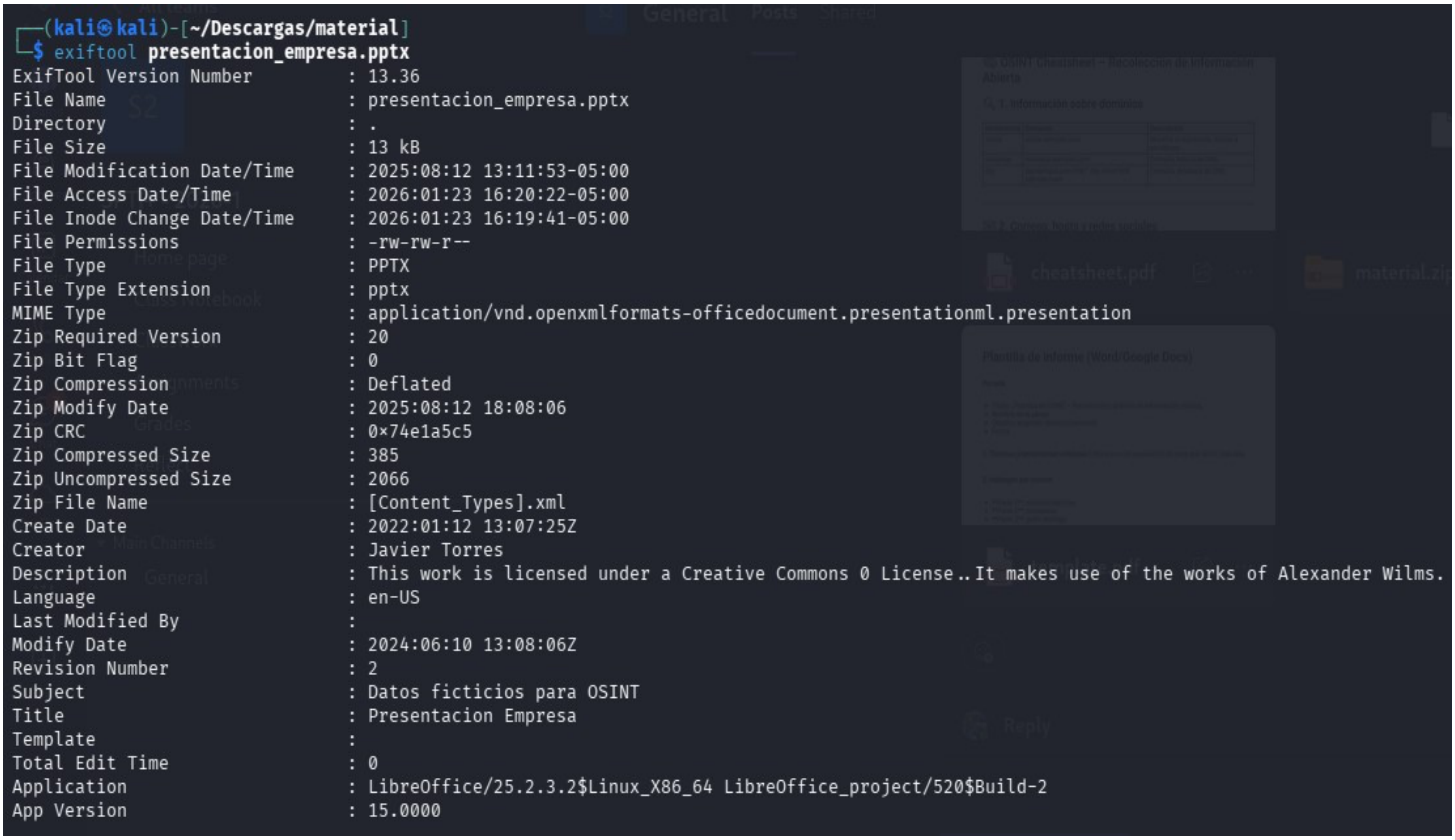
This image contains the highest level of exposure, as it includes full GPS metadata, allowing an exact geographical location to be determined. Personal data such as the author Mariana Gutiérrez, the equipment used Canon EOS 80D and the exact date and time of capture are also identified. The presence of coordinates makes this file the most sensitive of the set from the point of view of privacy and geolocation.

```
┌──(kali㉿kali)-[~/Descargas/material]
└─$ exiftool -gps:all *
======== contrato.docx
======== foto_evento.jpg
GPS Version ID                  : 2.3.0.0
GPS Latitude                    : 34 deg 36' 13.40"
GPS Longitude                   : 58 deg 22' 53.73"
======== informe_proyecto.pdf
======== logo_empresa.png
======== presentacion_empresa.pptx
    5 image files read
```




```
┌──(kali㉿kali)-[~/Descargas/material]
└─$ exiftool logo_empresa.png
ExifTool Version Number         : 13.36
File Name                       : logo_empresa.png
Directory                       : .
File Size                       : 12 kB
File Modification Date/Time     : 2025:08:12 13:51:43-05:00
File Access Date/Time           : 2026:01:23 16:19:43-05:00
File Inode Change Date/Time     : 2026:01:23 16:19:41-05:00
File Permissions                : -rw-rw-r--
File Type                       : PNG
File Type Extension             : png
MIME Type                       : image/png
Image Width                     : 300
Image Height                    : 240
Bit Depth                       : 8
Color Type                      : RGB with Alpha
Compression                     : Deflate/Inflate
Filter                          : Adaptive
Interlace                       : Noninterlaced
Background Color                : 255 255 255
Artist                          : Sofía Herrera
Software                        : Adobe Illustrator CC
Image Size                      : 300×240
Megapixels                      : 0.072
```

The file reveals the name Sofía Herrera in the Artist field and the use of Adobe Illustrator CC, medium sensitivity information that allows identifying authorship and the design software used. It does not contain detailed temporal data or geolocation, so the impact is limited to personal and technical attribution.




This file identifies Javier Torres as the creator and contains descriptive information that mentions the use of Creative Commons licensed material, as well as a purpose related to fictitious data for OSINT.

## 3. Part 3: Visualization in Maltego CE.

**Domain:** tesla.com
**Company:** Tesla
**Website:** https://www.tesla.com

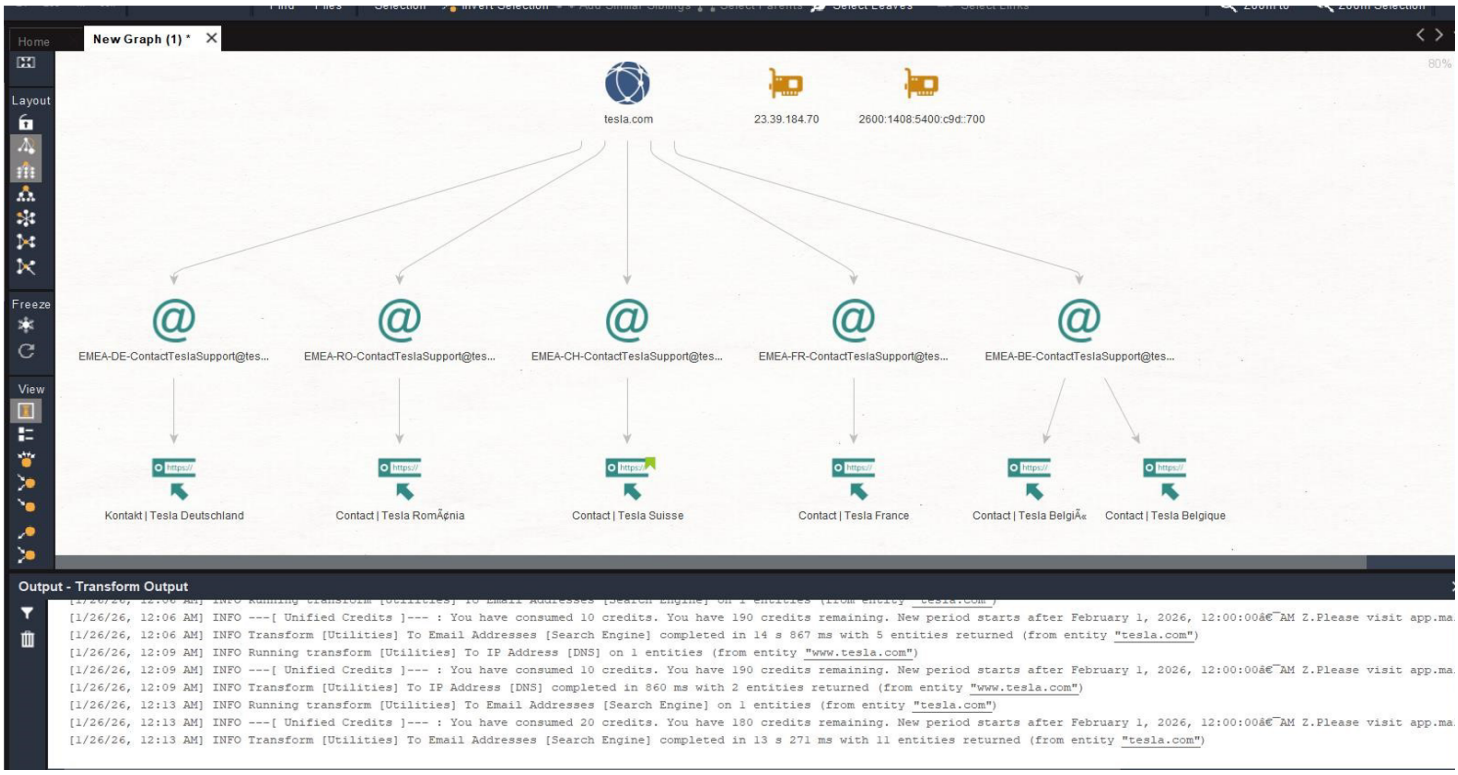



*Image 1.1. Searching for Dominion from Maltego*

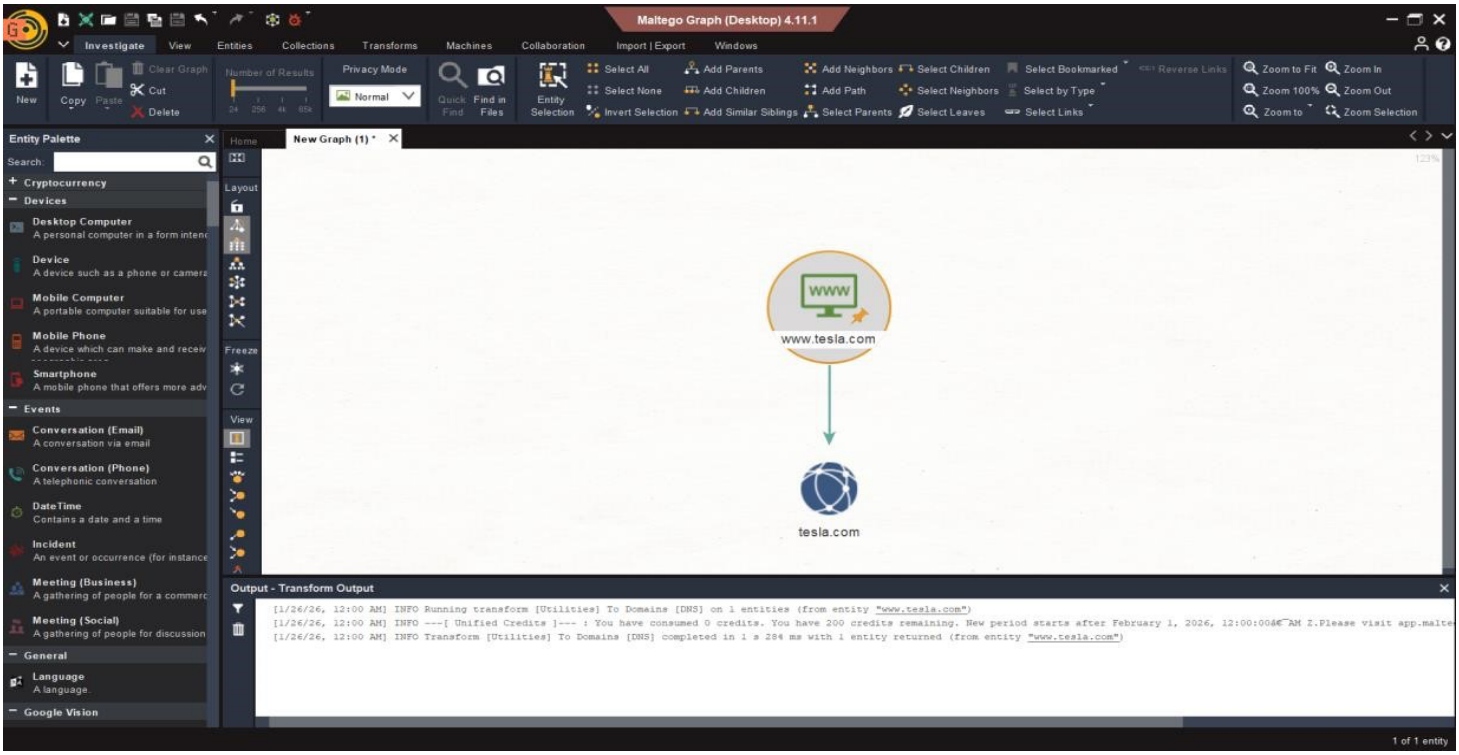



*Image 1.2. Defend Maltego*

- Some of the associated emails support Associated mail to countries and direct contact:
    - IP with which it gives us access and without the permissions we could access sensitive data.

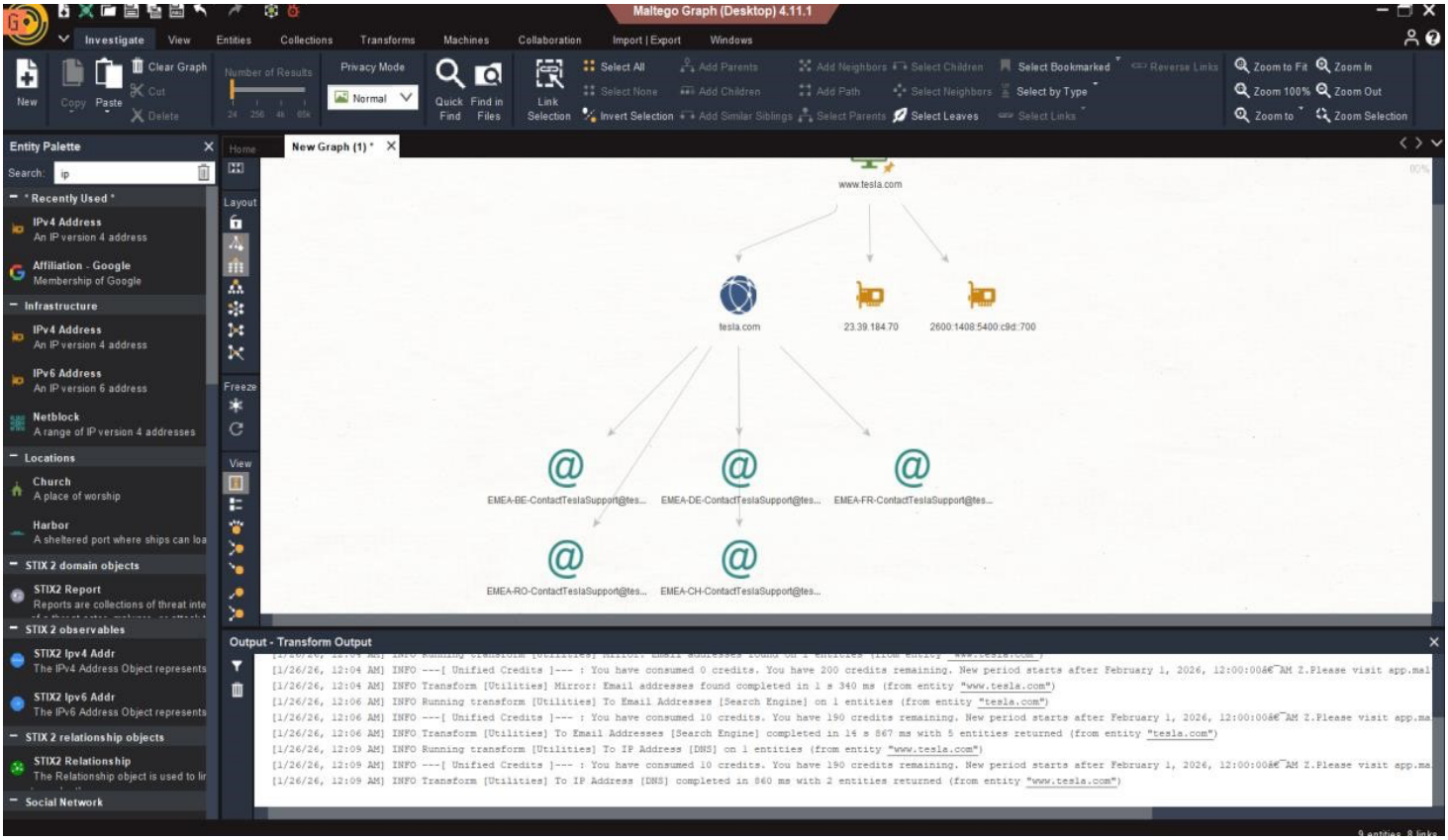
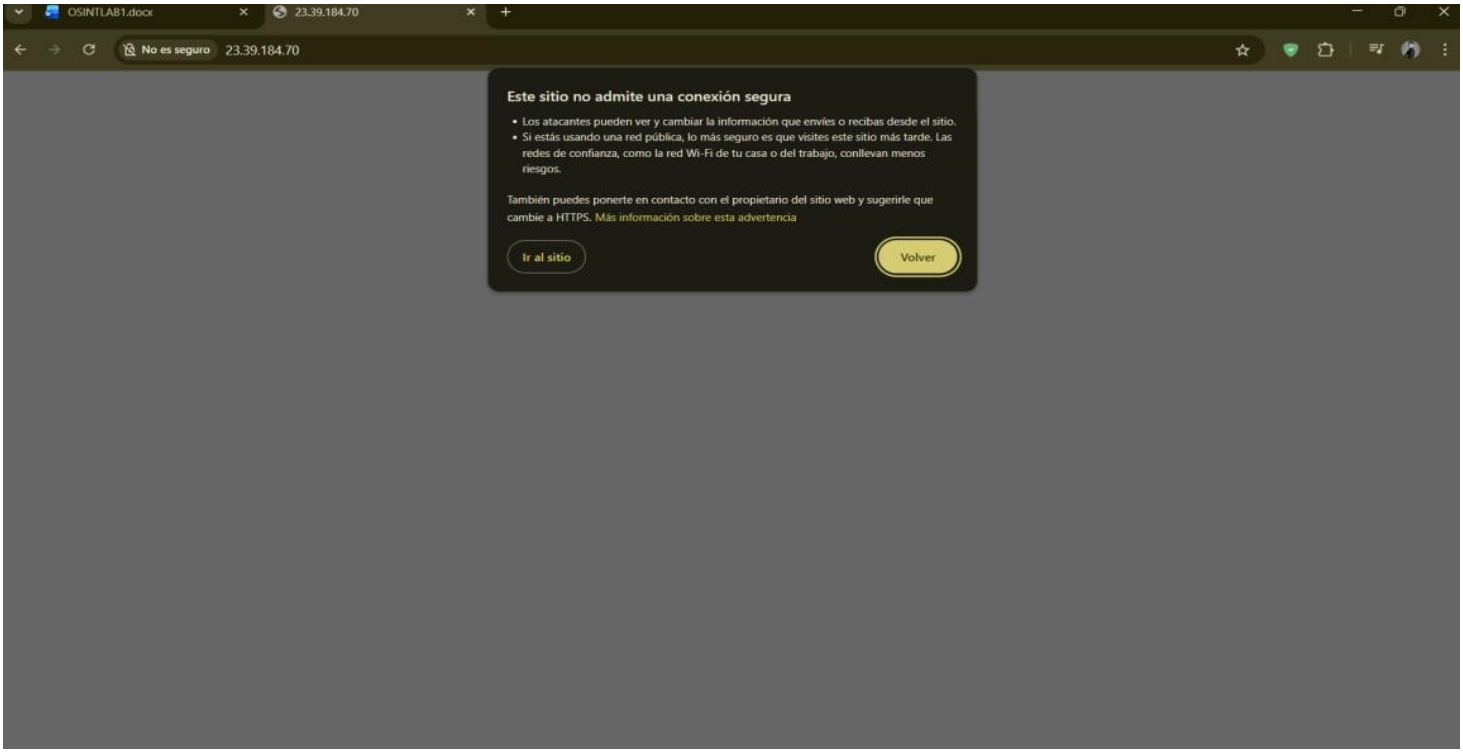



*Figure 1.3. Def Maltego Ip Access*



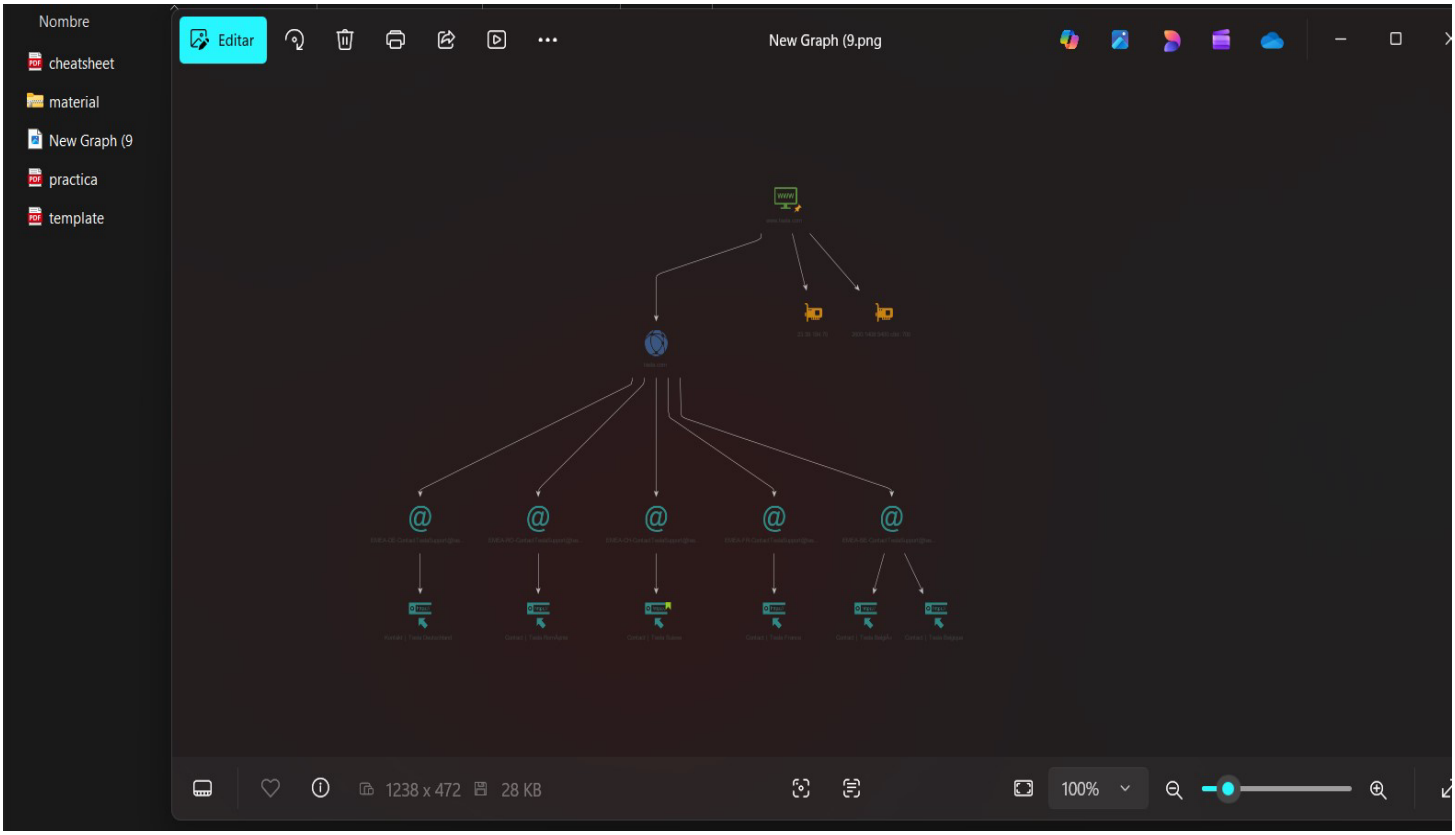



*Image 2.1. Final graph with information*

**4. Criterion Analysis**

1. Risks
    a. Our analysis shows that, although Tesla applies good perimeter security practices (domain protections, use of Akamai and UltraDNS, DNS balancing, WHOIS cloaking), the public exposure surface is large due to the large number of identified subdomains (more than 300 via Certificate Transparency)
    b. Concrete and verifiable risk: Each public subdomain increases the potential attack surface, especially if one belongs to staging environments. APIs. We clarify that no vulnerabilities are identified, but the volume of subdomains increases operational risk, something recognized in offensive and defensive security.
2. Ethics
    a. From an ethical point of view, the analysis is legal because it uses only public information, but it has weaknesses in not applying principles of minimization and anonymization. We expose personally identifiable data that were not strictly necessary to demonstrate the OSINT methodology, prioritizing technical demonstration over the protection of the individuals involved. In addition, there is no explicit statement of responsible use or ethical considerations about the impact of the analysis, which opens up the possibility of correlated information being reused outside of an academic context.

3. Privacy
    a. We evidenced significant exposure of personal data through persistent metadata in documents and images. Real names, organizational information, and precise geolocation data are identified, directly compromising the privacy of the individuals mentioned.
    b. The absence of measures such as metadata cleansing or anonymization demonstrates a lack of application of privacy principles by design. It cannot be confirmed that the identified individuals have consented to the use and analysis of their information, reinforcing the negative impact on privacy.

**5. Reflection**

1. As I analyzed this work, I realized how easy it is to get information from the internet without people knowing. Even though the data is public, that doesn't mean people are aware that others can see and use it. This made me think that posting files or photos carelessly can bring real problems, even without bad intentions.
2. I also understood that not everything that is legal is necessarily correct. During the analysis, real data from people was used, and although the goal was academic, I felt that more care could have been taken with their names and locations. This made me reflect on the importance of respecting others when working with information that is not ours.
3. This exercise helped me become more aware of my own privacy. Now I know that small things, like a photo or a document, can say a lot more than you might imagine. From this, I consider it important to be more responsible with what I share and learn to better protect my personal information.